

AMIT KUMAR

+91-7209279655 | amitr01032003@gmail.com | Leetcode | Github | LinkedIn

Education

Dr. APJ Abdul Kalam Technical University (A.K.G.E.C Ghaziabad)

(2023 – 2027)

Bachelor of Technology in Computer Science

Experience

Backend & Systems Development Trainee – BDCOE

(Oct-Dec)2024

- Designed and developed RESTful APIs with integrated authentication and RBAC across 3+ modules, ensuring secure and controlled access
- Implemented efficient data storage and retrieval using MongoDB, optimizing schema design for performance and scalability
- Improved API performance by optimizing MongoDB queries and implementing indexing, reducing response time
- Tested and validated APIs using Postman, ensuring reliability, performance, and correct endpoint behavior
- Diagnosed and resolved backend failures, API inconsistencies, and database issues, improving overall system robustness
- Enhanced application stability by identifying and fixing logic errors and integration bugs across services

Projects

Finance Tracker Backend |

- Designed and developed **15+ RESTful APIs** for financial record management, filtering, and analytics
- Implemented secure **JWT-based** authentication and Role-Based Access Control (**RBAC**) with **3 user roles** (Admin, Analyst, Viewer)
- Engineered efficient data handling using **MySQL and TypeORM**, ensuring reliable storage and retrieval of financial transactions
- Optimized dashboard aggregation APIs, achieving response times under **200ms** for real-time insights
- Architected a modular and scalable system with **5+ independent** modules following clean layered architecture (Controller–Service–Repository)

FLIGHT BOOKING SYSTEM – BACKEND MICROSERVICES

- Developed a real-world flight booking system using **Node.js** and **MySQL**, supporting seat booking, payments, and cancellations within a **microservices architecture**
- Resolved **concurrency** and **overbooking** challenges by implementing seat locking, controlled updates, and transactional workflows
- Built **idempotent payment** handling using **Redis**, preventing duplicate bookings and ensuring safe retry mechanisms
- Automated booking expiration using **cron jobs**, releasing reserved seats if payment is not completed within a defined time window
- Designed a robust booking lifecycle (INITIATED – BOOKED / CANCELLED) with strict validation, consistency checks, and rollback mechanisms
- Designed and tested scalable **RESTful APIs** for Flights, Airports, Airplanes, Cities, and Bookings, ensuring high data consistency
- Documented APIs using **Swagger** (OpenAPI) to enable structured documentation and seamless endpoint testing

Technical Skills

Languages : Python, JavaScript, C++

Technologies/Frameworks : Node.js, Express.js, RESTful APIs, Microservices

Developer Tools : Postman, Git(version control), MongoDB, Redis Docker, Kubernetes, Sequelize ORM, CI/CD

CS Fundamentals : Operating System, DBMS, Networking, DSA

Soft Skills : Team Leadership, Cross-Functional Collaboration, Project Management, Communication

Achievement

- Secured **3rd Place** at **Pradhyut Hackathon**, building an SDG-aligned solution within 24 hours, demonstrating innovation, rapid problem-solving, and teamwork
- Selected among top teams from AKGEC for **Smart India Hackathon (SIH)** Internal Round, qualifying to represent the institute at the national-level competition